Is Dale's law the implicit stabiliser?

exp062 · 11 July 2026 · Draft

Abstract

The claim: Dale's law is the implicit stabiliser that keeps a recurrent excitatory/inhibitory (E/I) spiking network trainable on the Spiking Heidelberg Digits (SHD) spoken-digit classification benchmark. Dale's law is the biological constraint that a neuron is either excitatory or inhibitory, so its outgoing synaptic weights all keep a fixed sign. Enforced during training as a non-negativity projection on the recurrent weights, it bounds those weights at every step. The question is whether that bound is what separates a net that trains to completion from one whose loss overflows to NaN (not-a-number, the signature of numerical divergence).

Two otherwise-identical cells, free (signed recurrence) versus Dale's law, run at the coarse timestep where the free net is known to overflow. The constraint removes the divergence outright: the free net loses 13 of 30 epochs to NaN, Dale's law 0, and best test accuracy barely moves. Dale's law is the implicit stabiliser.

Methods

exp061 killed the first hypothesis: the free signed-recurrent net's NaN divergence is not exponential-Euler stiffness at a coarse integration timestep Δt. Finer Δt made the pre-clip gradient explosion *worse*, because quartering Δt quadruples the backpropagation-through-time (BPTT) unroll and the recurrent gradient chain blows up. So the divergence is a property of the **free recurrence** itself, not the timestep. The plan's next candidate is the one structural constraint the free net drops: **Dale's law.** Its projection clamps the recurrent weights non-negative (and fixes their sign), which bounds the gain of the E→I→E (excitatory→inhibitory→excitatory) recurrent loop, plausibly below the runaway threshold the free net crosses.

The test, step by step:

1. **Two cells, one difference.** Hold every parameter fixed except the constraint: **free** (signed weights, the exp060/exp061 setting that diverges) versus **constrained** (Dale's law).
2. **The coarse-Δt regime.** Run both at the timestep Δt = 1 ms where the free net is known to overflow to NaN (exp061 here 13 of 30 epochs), at a single seed and full exp060 scale.
3. **The registered metrics.** Measure the plan's free-versus-constrained contrast: the NaN-epoch rate (the fraction of the run's training epochs whose loss is non-finite), the peak norm of the recurrent excitatory-to-excitatory weight matrix ‖W_ee‖, and the best test accuracy.

Parameter	Value
Dataset	Spiking Heidelberg Digits (SHD), 1000-sample subset
Model	COBANet (conductance-based E/I spiking network), 256 hidden units, all four recurrent weight blocks trainable
Contrast	free (*–no-dales-law*, signed) vs constrained (*–dales-law*, projected non-negative)
Integration	$\Delta t = 1$ ms, trial duration $T = 1000$ ms (the setting where the free net diverges)
Seeds	1 (seed 42), a quick single-seed contrast
Training	30 epochs, learning rate 0.001, weight-decay 0.001, batch size 32, surrogate-gradient backpropagation through time
Regulariser	upper firing-rate bound (threshold $\theta_u = 100$, strength $s_u = 0.06$)

Compute	runpod

Dale's law is the **only** thing that differs between the two cells, so any gap in the stability metrics is attributable to the constraint alone. Both cells run as a RunPod fan-out, one pod per cell.

> **Kill criterion.** If the Dale's-law net also overflows to NaN at matched settings, the constraint is not what stabilises: the difference lies elsewhere.

Results

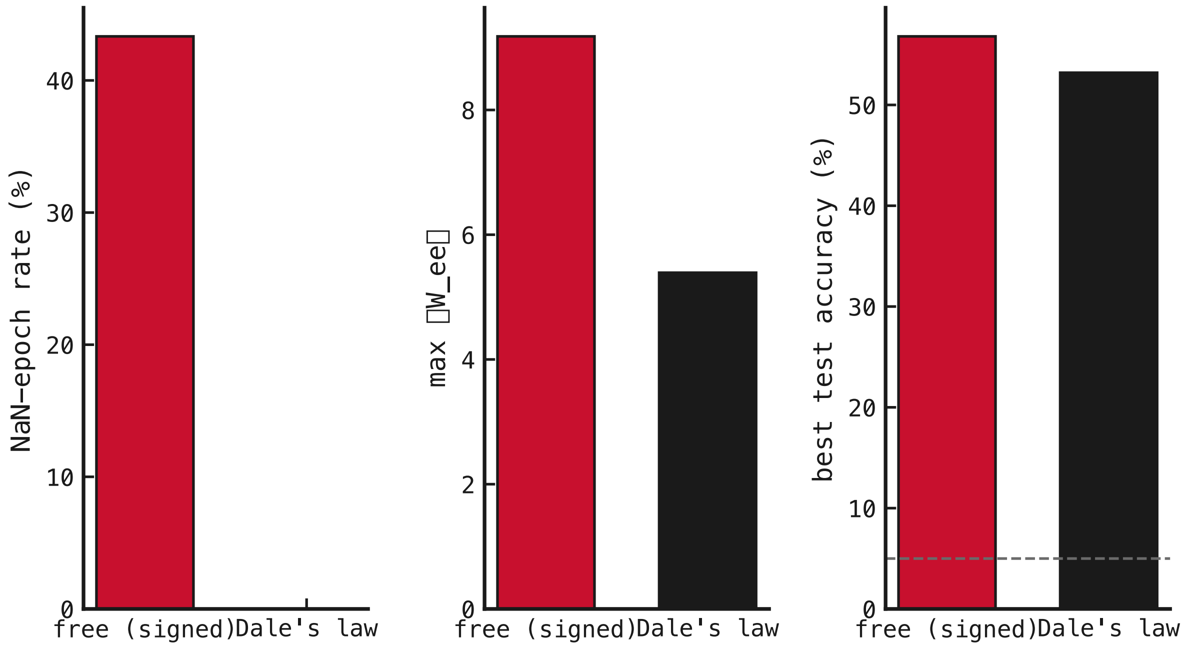

Figure 1: Free (red) versus Dale's law (black) across the three registered stability metrics, one bar pair per panel. Left to right: (a) NaN-epoch rate (percent of training epochs with non-finite loss); (b) peak norm of the recurrent excitatory-to-excitatory weight matrix, ‖W_ee‖ (dimensionless); (c) best test accuracy (percent), with the dashed line marking the 5% chance level for the 20 SHD classes. **What we see confirms it.** The free net (red) carries 13 of 30 epochs NaN and a peak ‖W_ee‖ of 9.2 Dale's law (black) drops the NaN-epoch rate to **zero** and holds ‖W_ee‖ at 5.4, while best accuracy barely moves (56.8% → 53.2%). The constraint buys stability for a few points of accuracy.

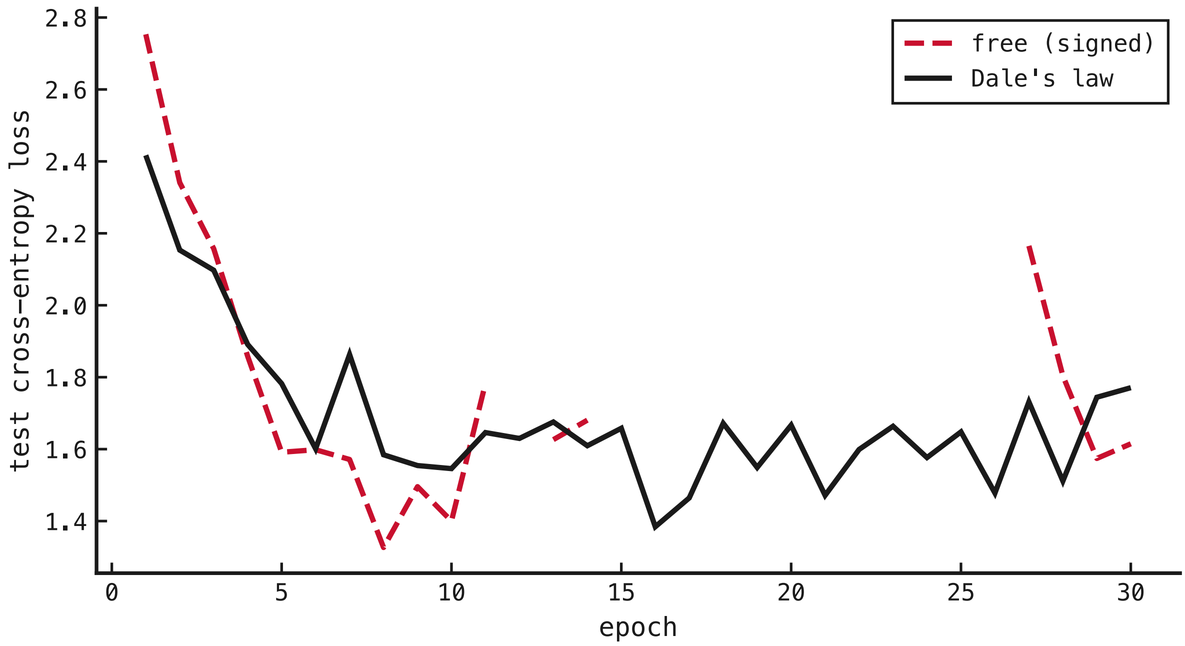

Figure 2: Per-epoch test cross-entropy loss, free (red) versus Dale's law (black). The horizontal axis is the training epoch; the vertical axis is test cross-entropy loss. NaN epochs (those whose loss is non-finite) appear as gaps in the curve: the free net leaves holes where its loss overflows, while the constrained net runs continuously to completion.

The free net diverges exactly as before: 13 of 30 epochs go NaN, with a peak pre-clip gradient norm of $\approx 10^4$, reproducing exp061′s coarse-Δt cell. Flip the single flag to Dale's law and, at otherwise identical settings, the divergence is **gone**: 0 NaN epochs, the peak gradient norm bounded at $\approx$ 190 (some 2 orders of magnitude tamer), and ‖W_ee‖ held at 5.4 against the free net's 9.2.

The hypothesis holds. Dale's law is the implicit stabiliser. Its non-negativity projection clamps the recurrent weights each step, bounding the E→I→E loop gain below the runaway threshold the free signed net crosses. The divergence never starts, so there is no gradient explosion to clip. This is the first **stable recipe** the program's goal asked for: a conductance-based E/I net that trains to completion on SHD with no NaN epochs and bounded dynamics, decisively above chance (53.2% against the 5% chance floor).

The cost, and where this points. Stability is not free: the constrained net gives up a few points of peak accuracy (53.2% vs 56.8%), the price of denying the signed recurrence its full expressivity. So two threads open. For the *minimal stable recipe*, Dale's law now answers it; the queue's remaining probe (exp063, weight decay) and the reserved state clamp become alternative stabilisers that may not be needed. For *recovering the free net's accuracy without its divergence*, those same tools become the live question: can weight decay or a forward-pass state clamp tame the signed net and keep its extra accuracy?

Caveat. Single seed, a point contrast, not error-barred. The effect is categorical (13 vs 0 NaN epochs), so a seed sweep is unlikely to reverse it; the confirming 3-seed run is cheap.